Chapter 1: Coupled Simulation

Now we couple locally mass conserved MFEM and FVM method together in this chapter. We discuss implementation details and coupling strategy combining MFEM, FVM and eutectic phase package. The remaining uncertainty of porosity ϕ will be fulfilled by phase package evaluation. Before we throw ourselves into the fully coupled problem. We test another porosity evolution scenerio with a more straightforward prescribed melting. Then we will show an example combining MFEM, FVM and eutectic phase package alltogether and perform coupled computation.

1.1 Coupled Algorithm

Let $U^n = (C^n, H^n)$ be the solution to the transport system at time t^n. Let $V^n = (\check{\mathbf{v}}^n_{h,r}, \check{\mathbf{v}}^n_{h,s}, \check{q}^n_{h,l}, \check{q}^n_h)$ denote the solution to the Darcy-Stokes system at the same time. Moreover, let $W^n = (\phi^n, \check{T}^n, c^n_s, c^n_l, \phi^n_1)$ denote the output of the phase calculation at t^n. We formulate an overall solution methods. The coupled solver is

Algorithm 1 Sequential Coupled Solver

Let U^0 be given as initial data.

1: **for** Time level $n = 0, 1, \ldots, n_{\max} - 1$ **do**

2: Given U^n, compute W^n by the eutectic phase package.

3: Given $\phi^n_f \in W^n$, solve the Darcy-Stokes system (**??**)–(**??**) for V^{n+1}.

4: Given W^n and V^{n+1}, compute effective velocity (**??**) and phase averaged velocity as

$$\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}, \quad \overline{\mathbf{v}} = \phi \mathbf{v}_r + \mathbf{v}_s.$$

5: Time stepping the transport system (**??**) to next time $n+1$ for U^{n+1}.

6: **end for**

initialized from the initial condition on the transport variables U^0 specifying the total initial mass and energy. The eutectic phase package gives W^0. Once we know W^0, we have phase related variables porosity and individual concentrations.

In Algorithm 1, U^{n+1} is computed using fixed in time values for the effective velocities $\mathbf{v}_{\text{eff}}$ and phase averaged velocity $\overline{\mathbf{v}}$, pre-calculated from the Darcy-Stokes part, and the partition functions κ and λ. These quantities are only updated before commencing the next time step.

1.1.1 Boundary conditions

In practical implementation, we prescribe boundary conditions to each dofs on the boundary. Each dofs will be associated with two types of boundary conditions.

1.2 Column Tests

We first test our coupled algorithm with a pseudo-1D column setting.

1.2.1 Prescribed Melting

Before fully coupled computation, we illustrate a simpler model coupling case coupling MFEM and FVM solver. Instead of deciding porosity with phase package computation, we prescribe the melting rate or the change of porosity as a fixed function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.1}$$

We write conservation of liquid mass with the prescribed melting term Γ (1.1) as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.2}$$

where diffusion effect is being dropped for simplicity.

We perform a quasi-1D column test on a 2×40 rectangular physical domain $\Omega \in [-0.05, 0.05] \times [-0.4, 0]$. We prescribe the following boundary conditions for MFEM and FVM solver.

- Bottom side;
  - MFEM. We prescirbe solid velocity at the bottom of the physical domain, indicating supplying of mantle.

    $$\check{\mathbf{v}}_s = \mathbf{v}_{\text{upwelling}}.$$

    We prescribe zero liquid velocity indicating no melting penetrating bottom of the computational domain;
  - FVM. We prescribe a fixed inflow boundary condition, supplying composition and enthalpy into system.
- Left and right side;
  - MFEM. We prescribe natural boundary conditon giving zero traction to the Stokes equation in tangential direction. We prescribe zero flux for dofs in normal direction for the Stokes equation and Darcy equation.
  - FVM. We prescribe noflow (zero flux) condition.
- Top side;
  - MFEM. We prescribe natural boundary conditoin given zero traction and zero pressure to Stokes and Darcy part respectively.
  - FVM. We set free outflow boundary if it is recognized as outflow. We set zero if the velocity reversed to inflow boundary condition.

For simplicity and efficiency in computation, we use 1×3 and 1×2 for our ML-WENO(3,2) stencils, since we don't need resolution and high order accuracy in x direction. We show two different melting source scenarios. One of which is a piece wise constant stair function:

$$\Gamma_1(z) = \begin{cases} 1e-3, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \tag{1.3}$$

and one with melting source trapped in the middle of the melting column as

$$\Gamma_2(z) = \begin{cases} 0, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \tag{1.4}$$

We assign different initial conditions to these two scenarios as

$$\phi_1(z,0) = 0.0, \quad z \in [-0.4, 0.0], \tag{1.5}$$

and

$$\phi_2(z,0) = \begin{cases} 1e-2, & -0.2 < z \leq 0, \\ 0, & -0.4 \leq z \leq -0.2. \end{cases} \tag{1.6}$$

1.2.2 Phase Transition

1.3 Two-Dimensional Case